Class 10:

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Table of contents

Background

The main repository of high-resolution structural data on biomolecules is called the **Protein Data Bank** (PDB)

PDB statistics

What is in the PDB in terms of molecule type and structure determination method?

Read a CSV file of current PDB stats from: https://www.rcsb.org/stats/summary

```
pdb <- read.csv ("Data Export Summary.csv")
pdb
```

```
           Molecular.Type   X.ray      EM     NMR Integrative Multiple.methods
1          Protein (only) 180,758 23,111 12,813         348              229
2 Protein/Oligosaccharide  10,488  3,741     34           8               11
3              Protein/NA   9,205  6,751    287          26                8
4     Nucleic acid (only)   3,154    250  1,578           3               15
5                   Other     178     27     35           4                0
6  Oligosaccharide (only)      11      0      6           0                1
  Neutron Other   Total
1      84    32 217,375
2       1     0  14,283
3       0     0  16,277
4       3     1   5,004
5       0     0     244
6       0     4      22
```

```
pdb$X.ray
```

```
[1] "180,758" "10,488"  "9,205"   "3,154"   "178"     "11"
```

This print out above **pdb$X.ray** is a "character" not "numeric". Therefore I can't do math with it. We need to fix this...

Two functions that can help `sub()` and `as.numeric()`

```
# We want to get rid (or sub out) commas:

x <- pdb$X.ray
tmp <- sub(",","",x)
sum(as.numeric(tmp))
```

```
[1] 203794
```

We could make a function to do this:

```
rm.comma <- function(x) {
  tmp <- sub(",","",x)
sum(as.numeric(tmp)) }
```

```
rm.comma(pdb$X.ray)
```

```
[1] 203794
```

```r
rm.comma(pdb$EM)
```

```
[1] 33880
```

We could also use a different import function for this CSV that speaks American (i.e. deals with commas in numbers in a comma separated value file).

> Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

So for X-ray, we got 80.5% and for EM, it was 13.4%.

```r
library(readr)
pdb <- read_csv("Data Export Summary.csv")
```

```
Rows: 6 Columns: 9
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (1): Molecular Type
dbl (4): Integrative, Multiple methods, Neutron, Other
num (4): X-ray, EM, NMR, Total

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
n.tot <- sum(pdb$Total)
n.xray <- sum(pdb$`X-ray`)
n.em <- sum(pdb$EM)

n.xray / n.tot * 100
```

```
[1] 80.48577
```

```r
n.em / n.tot * 100
```

```
[1] 13.38046
```

```
sum(pdb$`Xray`) / n.tot
```

```
Warning: Unknown or uninitialised column: `Xray`.
```

```
[1] 0
```

So for X-ray, we got 80.5% and for EM, it was 13.4%. > **Key-point**: We have a very, very small structural coverage of known proteins (~0.1%). Most structures we know about (~80%) come from one method (X-ray Crystalography).

> Q2. What proportion of structures in the PDB are protein?

~85%of structures in PDB are Protein.

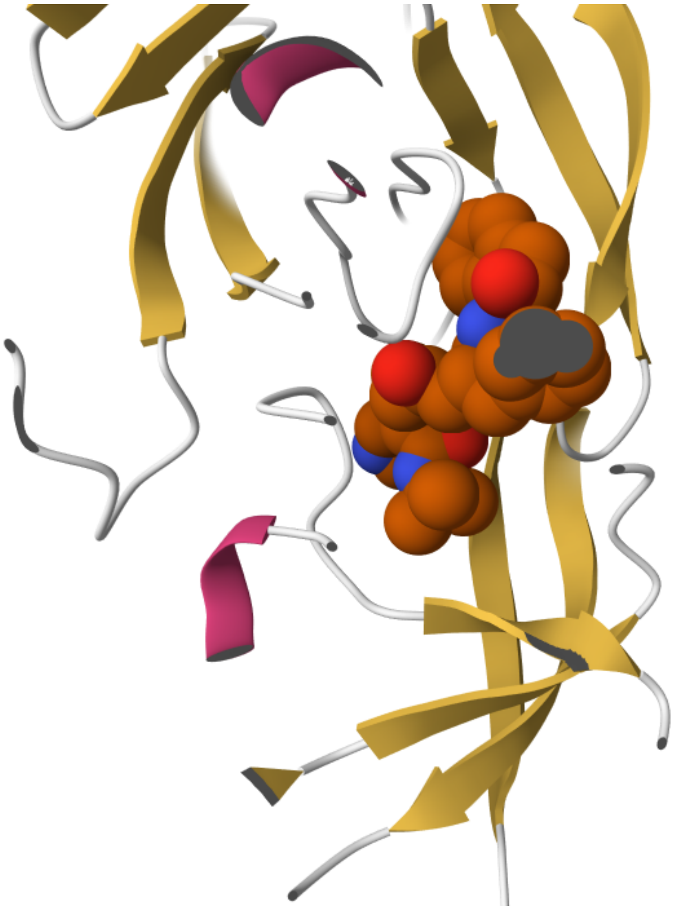
```
n.protein <- sum(pdb$Total[pdb$`Molecular Type` == "Protein (only)"])
n.protein / n.tot *100
```

```
[1] 85.84941
```

> Q3. Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB?

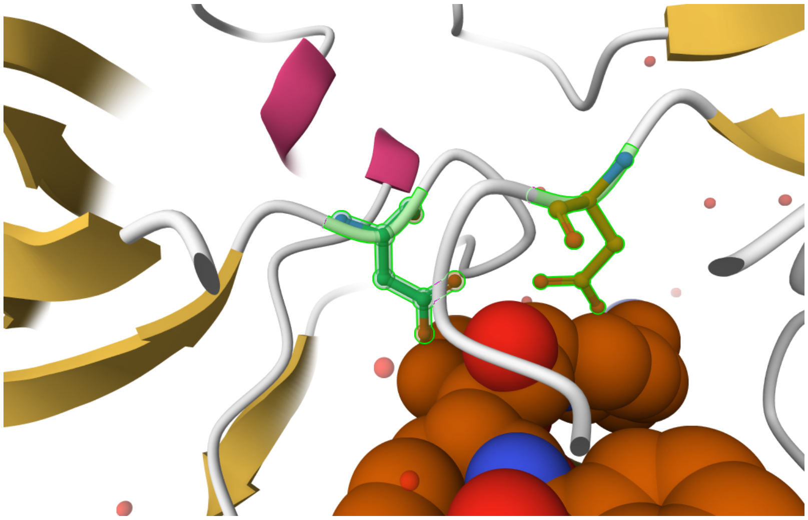
1,147 Structures

Visualizing the HIV-1 Protease Structure



> Q4: Water molecules normally have 3 atoms. Why do we see just one atom per water molecule in this structure?

We only see one atom per water molecule because hydrogen atoms are too small to be detected by X-ray crystallography. Causing for us to only see the only one oxygen atom of each water molecule.



Q5: There is a critical "conserved" water molecule in the binding site. Can you identify this water molecule? What residue number does this water molecule have

HOH 13, is the water molecule that is in the active site as well as interacting with ASP25.

Q6: Generate and save a figure clearly showing the two distinct chains of HIV-protease along with the ligand. You might also consider showing the catalytic residues ASP 25 in each chain and the critical water (we recommend "Ball & Stick" for these side-chains). Add this figure to your Quarto document.

> Discussion Topic: Can you think of a way in which indinavir, or even larger ligands and substrates, could enter the binding site?

Larger ligands such as indinavir can enter the binding site because of HIV Protease flexibility. Since the enzyme has flap regions that open toexpose the active site and close when ligand binds. Because the active site lies between the two chains of the homodimer, these structural movents are essential for substrate access.

Introduction to Bio3D in R

```
library(bio3d)

pdb <- read.pdb("1hsg")
```

```
  Note: Accessing on-line PDB file
```

```
pdb
```

```
 Call:  read.pdb(file = "1hsg")

   Total Models#: 1
     Total Atoms#: 1686,  XYZs#: 5058  Chains#: 2  (values: A B)

     Protein Atoms#: 1514  (residues/Calpha atoms#: 198)
     Nucleic acid Atoms#: 0  (residues/phosphate atoms#: 0)
```

```
     Non-protein/nucleic Atoms#: 172  (residues: 128)
     Non-protein/nucleic resid values: [ HOH (127), MK1 (1) ]

   Protein sequence:
      PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
      QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
      ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
      VNIIGRNLLTQIGCTLNF

+ attr: atom, xyz, seqres, helix, sheet,
        calpha, remark, call
```

> Q7: How many amino acid residues are there in this pdb object?

198 amino acid residues are in the pdb object

> Q8: Name one of the two non-protein residues?

HOH, or MK1

> Q9: How many protein chains are in this structure?

There are 2 chains.

```r
attributes(pdb)
```

```
$names
[1] "atom"   "xyz"    "seqres" "helix"  "sheet"  "calpha" "remark" "call"

$class
[1] "pdb" "sse"
```

```r
head(pdb$atom)
```

```
  type eleno elety  alt resid chain resno insert      x      y     z o     b
1 ATOM     1     N <NA>   PRO     A     1   <NA> 29.361 39.686 5.862 1 38.10
2 ATOM     2    CA <NA>   PRO     A     1   <NA> 30.307 38.663 5.319 1 40.62
3 ATOM     3     C <NA>   PRO     A     1   <NA> 29.760 38.071 4.022 1 42.64
4 ATOM     4     O <NA>   PRO     A     1   <NA> 28.600 38.302 3.676 1 43.40
5 ATOM     5    CB <NA>   PRO     A     1   <NA> 30.508 37.541 6.342 1 37.87
6 ATOM     6    CG <NA>   PRO     A     1   <NA> 29.296 37.591 7.162 1 38.40
  segid elesy charge
```

```
1  <NA>     N   <NA>
2  <NA>     C   <NA>
3  <NA>     C   <NA>
4  <NA>     O   <NA>
5  <NA>     C   <NA>
6  <NA>     C   <NA>
```

Quick PDB visualization in R

```r
library(bio3dview)
library(NGLVieweR)

view.pdb(pdb) |>
  setSpin()
```

```r
# Select the important ASP 25 residue
sele <- atom.select(pdb, resno=25)

# and highlight them in spacefill representation
view.pdb(pdb, cols=c("red","blue"),
         highlight = sele,
         highlight.style = "spacefill", backgroundColor = "pink") |>
  setRock()
```

Predicting Functional Motions of a single structure

```r
adk <- read.pdb("6s36")
```

```
  Note: Accessing on-line PDB file
   PDB has ALT records, taking A only, rm.alt=TRUE
```

```r
adk
```

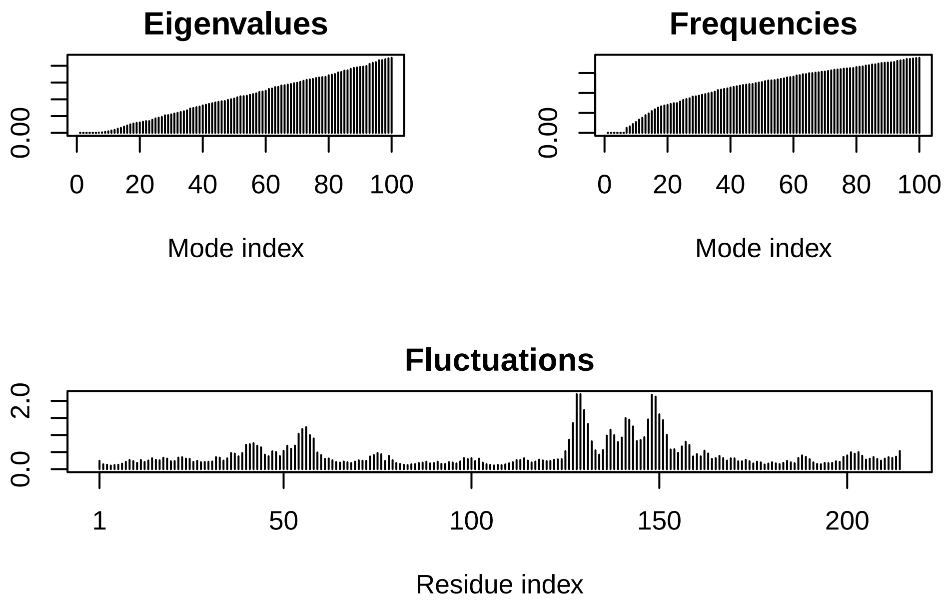
```
 Call:  read.pdb(file = "6s36")

   Total Models#: 1
```

```
     Total Atoms#: 1898,  XYZs#: 5694  Chains#: 1  (values: A)

     Protein Atoms#: 1654  (residues/Calpha atoms#: 214)
     Nucleic acid Atoms#: 0  (residues/phosphate atoms#: 0)

     Non-protein/nucleic Atoms#: 244  (residues: 244)
     Non-protein/nucleic resid values: [ CL (3), HOH (238), MG (2), NA (1) ]

   Protein sequence:
      MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLVT
      DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDKI
      VGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
      YYSKEAEAGNTKYAKVDGTKPVAEVRADLEKILG

+ attr: atom, xyz, seqres, helix, sheet,
        calpha, remark, call
```

```r
# Perform flexiblity prediction
m <- nma(adk)
```

```
 Building Hessian...        Done in 0.013 seconds.
 Diagonalizing Hessian...   Done in 0.271 seconds.
```

```r
plot(m)
```




```r
mktrj(m, file="adk_m7.pdb")
```

```r
view.nma(m, pdb=adk)
```

Write out the results for viewing in Mol - Star

Comparative Structure Analysis of Adenylate Kinase

> Q10. Which of the packages above is found only on BioConductor and not CRAN?

The only package that is only on Bioconductor is msa.

> Q11. Which of the above packages is not found on BioConductor or CRAN?:

bio3dview is not found on CRAN or BioConductor

> Q12. True or False? Functions from the pak package can be used to install packages from GitHub and BitBucket?

True.

```r
library(bio3d)
aa <- get.seq("1ake_A")
```

```
Warning in get.seq("1ake_A"): Removing existing file: seqs.fasta
```

```
Fetching... Please wait. Done.
```

```r
aa
```

```
             1        .         .         .         .         .         60
pdb|1AKE|A   MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLVT
             1        .         .         .         .         .         60

            61        .         .         .         .         .         120
pdb|1AKE|A   DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDRI
            61        .         .         .         .         .         120

           121        .         .         .         .         .         180
pdb|1AKE|A   VGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
           121        .         .         .         .         .         180

           181        .         .         .   214
pdb|1AKE|A   YYSKEAEAGNTKYAKVDGTKPVAEVRADLEKILG
           181        .         .         .   214

Call:
  read.fasta(file = outfile)

Class:
  fasta

Alignment dimensions:
  1 sequence rows; 214 position columns (214 non-gap, 0 gap)

+ attr: id, ali, call
```

> Q13. How many amino acids are in this sequence, i.e. how long is this sequence?

214 amino acids are in this sequence.

```r
hits <- NULL
hits$pdb.id <- c('1AKE_A','6S36_A','6RZE_A','3HPR_A','1E4V_A','5EJE_A','1E4Y_A','3X2S_A','6H
```

```r
files <- get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE)
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1AKE.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6S36.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6RZE.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3HPR.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4V.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/5EJE.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4Y.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3X2S.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAP.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAM.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4K46.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3GMT.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4PZL.pdb.gz exists. Skipping download
```

```
  |
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  |======================================================                |  77%
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  |===========================================================           |  85%
  |
  |=================================================================     |  92%
  |
  |======================================================================| 100%
```

```r
pdbs <- pdbaln(files, fit = TRUE, exefile = "msa")
```

```
Reading PDB files:
pdbs/split_chain/1AKE_A.pdb
pdbs/split_chain/6S36_A.pdb
```

```
pdbs/split_chain/6RZE_A.pdb
pdbs/split_chain/3HPR_A.pdb
pdbs/split_chain/1E4V_A.pdb
pdbs/split_chain/5EJE_A.pdb
pdbs/split_chain/1E4Y_A.pdb
pdbs/split_chain/3X2S_A.pdb
pdbs/split_chain/6HAP_A.pdb
pdbs/split_chain/6HAM_A.pdb
pdbs/split_chain/4K46_A.pdb
pdbs/split_chain/3GMT_A.pdb
pdbs/split_chain/4PZL_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..   PDB has ALT records, taking A only, rm.alt=TRUE
....   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
...

Extracting sequences

pdb/seq: 1   name: pdbs/split_chain/1AKE_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 2   name: pdbs/split_chain/6S36_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 3   name: pdbs/split_chain/6RZE_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 4   name: pdbs/split_chain/3HPR_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 5   name: pdbs/split_chain/1E4V_A.pdb
pdb/seq: 6   name: pdbs/split_chain/5EJE_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 7   name: pdbs/split_chain/1E4Y_A.pdb
pdb/seq: 8   name: pdbs/split_chain/3X2S_A.pdb
pdb/seq: 9   name: pdbs/split_chain/6HAP_A.pdb
pdb/seq: 10   name: pdbs/split_chain/6HAM_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 11   name: pdbs/split_chain/4K46_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 12   name: pdbs/split_chain/3GMT_A.pdb
pdb/seq: 13   name: pdbs/split_chain/4PZL_A.pdb
```

```r
library(bio3dview)
view.pdbs(pdbs)
```

Annotate Collected PDB structures

```r
# Vector containing PDB database codes
ids <- basename.pdb(pdbs$id)

anno <- pdb.annotate(ids)
unique(anno$source)
```

```
[1] "Escherichia coli"
[2] "Escherichia coli K-12"
[3] "Escherichia coli O139:H28 str. E24377A"
[4] "Escherichia coli str. K-12 substr. MDS42"
[5] "Photobacterium profundum"
[6] "Burkholderia pseudomallei 1710b"
[7] "Francisella tularensis subsp. tularensis SCHU S4"
```

```r
anno
```

```
       structureId chainId macromoleculeType chainLength experimentalTechnique
1AKE_A        1AKE       A           Protein         214                    X-
ray
6S36_A        6S36       A           Protein         214                    X-
ray
6RZE_A        6RZE       A           Protein         214                    X-
ray
3HPR_A        3HPR       A           Protein         214                    X-
ray
1E4V_A        1E4V       A           Protein         214                    X-
ray
5EJE_A        5EJE       A           Protein         214                    X-
ray
1E4Y_A        1E4Y       A           Protein         214                    X-
ray
3X2S_A        3X2S       A           Protein         214                    X-
ray
```

```
6HAP_A        6HAP       A           Protein         214                 X-
ray
6HAM_A        6HAM       A           Protein         214                 X-
ray
4K46_A        4K46       A           Protein         214                 X-
ray
3GMT_A        3GMT       A           Protein         230                 X-
ray
4PZL_A        4PZL       A           Protein         242                 X-
ray
       resolution       scopDomain                                       pfam
1AKE_A       2.00 Adenylate kinase                     Adenylate kinase (ADK)
6S36_A       1.60             <NA>                     Adenylate kinase (ADK)
6RZE_A       1.69             <NA>                     Adenylate kinase (ADK)
3HPR_A       2.00             <NA>                     Adenylate kinase (ADK)
1E4V_A       1.85 Adenylate kinase                     Adenylate kinase (ADK)
5EJE_A       1.90             <NA>                     Adenylate kinase (ADK)
1E4Y_A       1.85 Adenylate kinase                     Adenylate kinase (ADK)
3X2S_A       2.80             <NA>                                       <NA>
6HAP_A       2.70             <NA>                     Adenylate kinase (ADK)
6HAM_A       2.55             <NA>                     Adenylate kinase (ADK)
4K46_A       2.01             <NA>                     Adenylate kinase (ADK)
3GMT_A       2.10             <NA>                                       <NA>
4PZL_A       2.10             <NA> Adenylate kinase, active site lid (ADK_lid)
                ligandId
1AKE_A               AP5
6S36_A CL (3),MG (2),NA
6RZE_A    NA (3),CL (2)
3HPR_A               AP5
1E4V_A               AP5
5EJE_A            AP5,CO
1E4Y_A               AP5
3X2S_A   JPY (2),AP5,MG
6HAP_A               AP5
6HAM_A               AP5
4K46_A      ADP,AMP,PO4
3GMT_A          SO4 (2)
4PZL_A       FMT,GOL,CA
                                                                   ligandName
1AKE_A                                           BIS(ADENOSINE)-5'-
PENTAPHOSPHATE
6S36_A                              CHLORIDE ION (3),MAGNESIUM ION (2),SODIUM ION
6RZE_A                                         SODIUM ION (3),CHLORIDE ION (2)
```

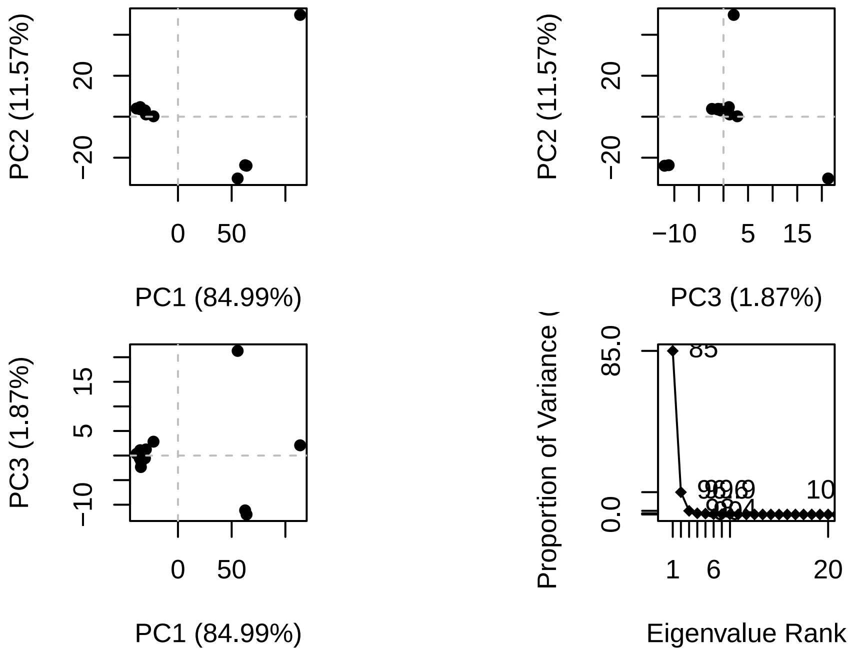
```
3HPR_A                                              BIS(ADENOSINE)-5'-
PENTAPHOSPHATE
1E4V_A                                              BIS(ADENOSINE)-5'-
PENTAPHOSPHATE
5EJE_A                                BIS(ADENOSINE)-5'-PENTAPHOSPHATE,COBALT (II) ION
1E4Y_A                                              BIS(ADENOSINE)-5'-
PENTAPHOSPHATE
3X2S_A N-(pyren-1-ylmethyl)acetamide (2),BIS(ADENOSINE)-5'-PENTAPHOSPHATE,MAGNESIUM ION
6HAP_A                                              BIS(ADENOSINE)-5'-
PENTAPHOSPHATE
6HAM_A                                              BIS(ADENOSINE)-5'-
PENTAPHOSPHATE
4K46_A                 ADENOSINE-5'-DIPHOSPHATE,ADENOSINE MONOPHOSPHATE,PHOSPHATE ION
3GMT_A                                                                SULFATE ION (2)
4PZL_A                                              FORMIC ACID,GLYCEROL,CALCIUM ION
                                              source
1AKE_A                                Escherichia coli
6S36_A                                Escherichia coli
6RZE_A                                Escherichia coli
3HPR_A                           Escherichia coli K-12
1E4V_A                                Escherichia coli
5EJE_A           Escherichia coli O139:H28 str. E24377A
1E4Y_A                                Escherichia coli
3X2S_A         Escherichia coli str. K-12 substr. MDS42
6HAP_A           Escherichia coli O139:H28 str. E24377A
6HAM_A                           Escherichia coli K-12
4K46_A                        Photobacterium profundum
3GMT_A                  Burkholderia pseudomallei 1710b
4PZL_A Francisella tularensis subsp. tularensis SCHU S4

1AKE_A STRUCTURE OF THE COMPLEX BETWEEN ADENYLATE KINASE FROM ESCHERICHIA COLI AND THE INHIBI
6S36_A
6RZE_A
3HPR_A
1E4V_A
loop
5EJE_A                                                                                 Crys
1E4Y_A
loop
3X2S_A
conjugated adenylate kinase
6HAP_A
6HAM_A
```

```
4K46_A
3GMT_A
4PZL_A                                                                              The crys
                                                            citation rObserved   rFree
1AKE_A              Muller, C.W., et al. J Mol Biology (1992)   0.19600      NA
6S36_A                  Rogne, P., et al. Biochemistry (2019)   0.16320 0.23560
6RZE_A                  Rogne, P., et al. Biochemistry (2019)   0.18650 0.23500
3HPR_A  Schrank, T.P., et al. Proc Natl Acad Sci U S A (2009)   0.21000 0.24320
1E4V_A                   Muller, C.W., et al. Proteins (1993)   0.19600      NA
5EJE_A  Kovermann, M., et al. Proc Natl Acad Sci U S A (2017)   0.18890 0.23580
1E4Y_A                   Muller, C.W., et al. Proteins (1993)   0.17800      NA
3X2S_A                 Fujii, A., et al. Bioconjug Chem (2015)   0.20700 0.25600
6HAP_A                Kantaev, R., et al. J Phys Chem B (2018)   0.22630 0.27760
6HAM_A                Kantaev, R., et al. J Phys Chem B (2018)   0.20511 0.24325
4K46_A                    Cho, Y.-J., et al. To be published    0.17000 0.22290
3GMT_A Buchko, G.W., et al. Biochem Biophys Res Commun (2010)   0.23800 0.29500
4PZL_A                       Tan, K., et al. To be published    0.19360 0.23680
         rWork spaceGroup
1AKE_A 0.19600  P 21 2 21
6S36_A 0.15940    C 1 2 1
6RZE_A 0.18190    C 1 2 1
3HPR_A 0.20620  P 21 21 2
1E4V_A 0.19600  P 21 2 21
5EJE_A 0.18630  P 21 2 21
1E4Y_A 0.17800   P 1 21 1
3X2S_A 0.20700 P 21 21 21
6HAP_A 0.22370    I 2 2 2
6HAM_A 0.20311       P 43
4K46_A 0.16730 P 21 21 21
3GMT_A 0.23500   P 1 21 1
4PZL_A 0.19130       P 32
```

Search ## Principal Component Analysis

```r
# Perform PCA
pc.xray <- pca(pdbs)
plot(pc.xray)
```

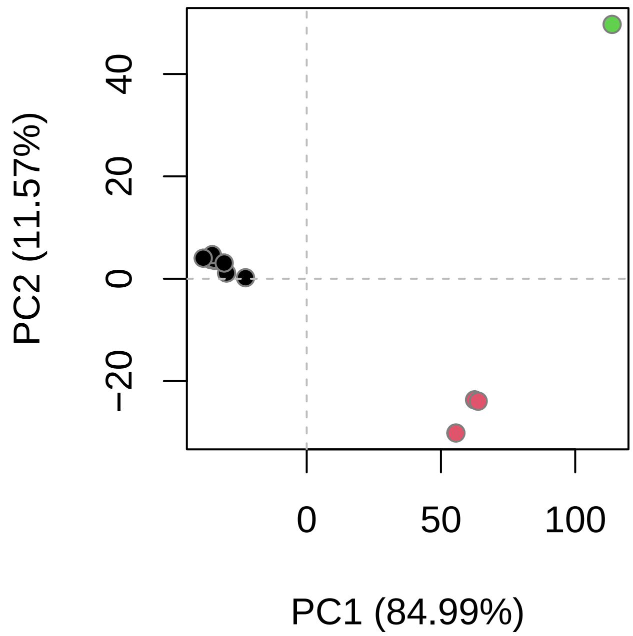



```
# Calculate RMSD
rd <- rmsd(pdbs)
```

```
Warning in rmsd(pdbs): No indices provided, using the 204 non NA positions
```

```
# Structure-based clustering
hc.rd <- hclust(dist(rd))
grps.rd <- cutree(hc.rd, k=3)

plot(pc.xray, 1:2, col="grey50", bg=grps.rd, pch=21, cex=1)
```



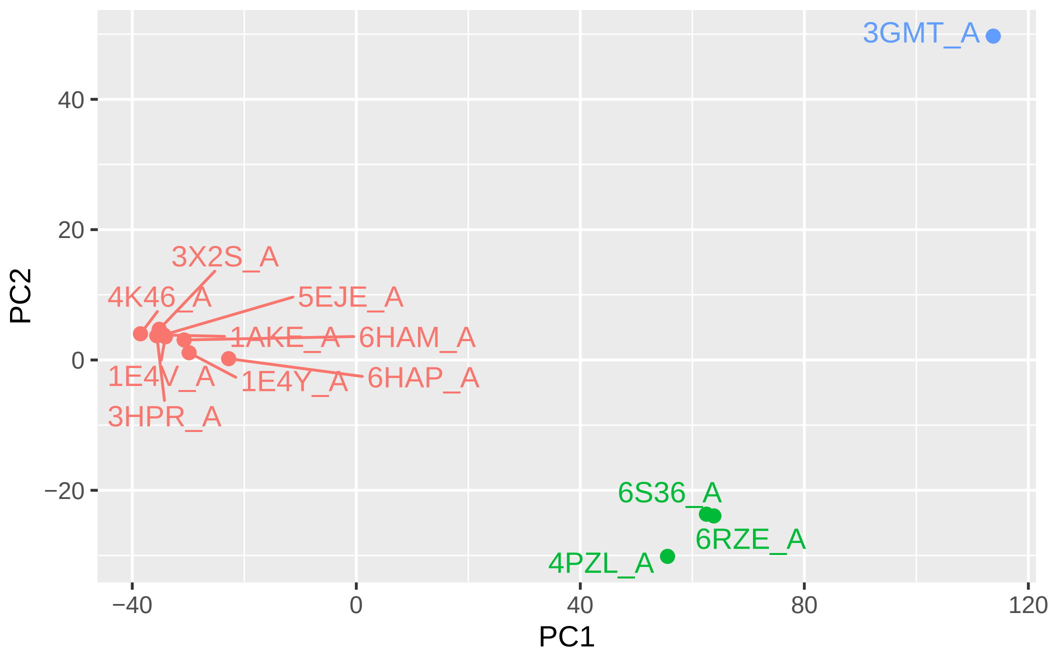
PCA Visualization

```
# Visualize first principal component
pc1 <- mktrj(pc.xray, pc=1, file="pc_1.pdb")
```

```
#Plotting results with ggplot2
library(ggplot2)
library(ggrepel)

df <- data.frame(PC1=pc.xray$z[,1],
                 PC2=pc.xray$z[,2],
                 col=as.factor(grps.rd),
                 ids=ids)

p <- ggplot(df) +
  aes(PC1, PC2, col=col, label=ids) +
  geom_point(size=2) +
  geom_text_repel(max.overlaps = 20) +
  theme(legend.position = "none")
p
```



Normal Mode Analysis

```
modes <- nma(pdbs)
```

```
Details of Scheduled Calculation:
  ... 13 input structures
  ... storing 606 eigenvectors for each structure
  ... dimension of x$U.subspace: ( 612x606x13 )
  ... coordinate superposition prior to NM calculation
  ... aligned eigenvectors (gap containing positions removed)
  ... estimated memory usage of final 'eNMA' object: 36.9 Mb
```

```
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```

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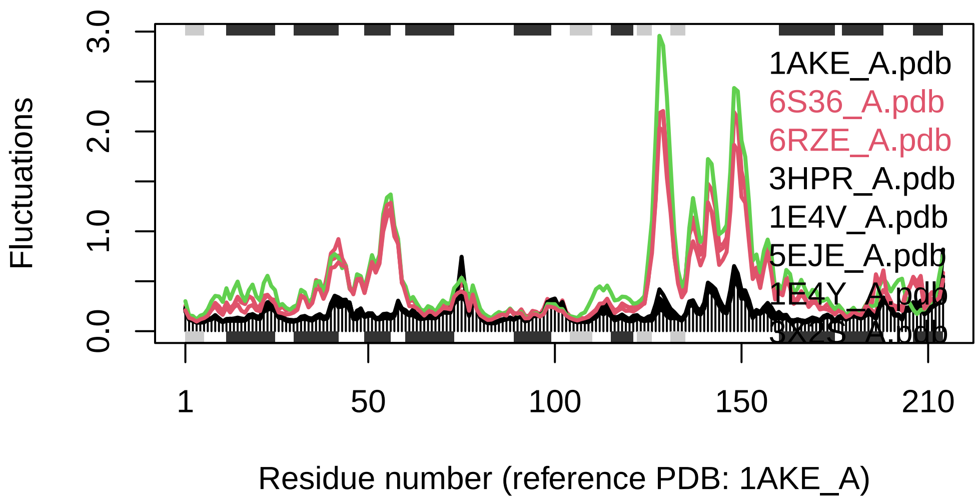
  |===========================================================           |  85%
  |
  |=================================================================     |  92%
  |
  |======================================================================| 100%
```

```r
plot(modes, pdbs, col=grps.rd)
```

```
Extracting SSE from pdbs$sse attribute
```



Q14. What do you note about this plot? Are the black and colored lines similar or different? Where do you think they differ most and why?

The black and colored lines are mostly similar, but around 120-160 is where the colored lines show high fluctuations. This indicates increased flexibility in these reigions, likely due to conformational changes in functional sites like the binding pocket.